

Server Deployment with NAS Storage

Project Title: Server Deployment & NAS Integration

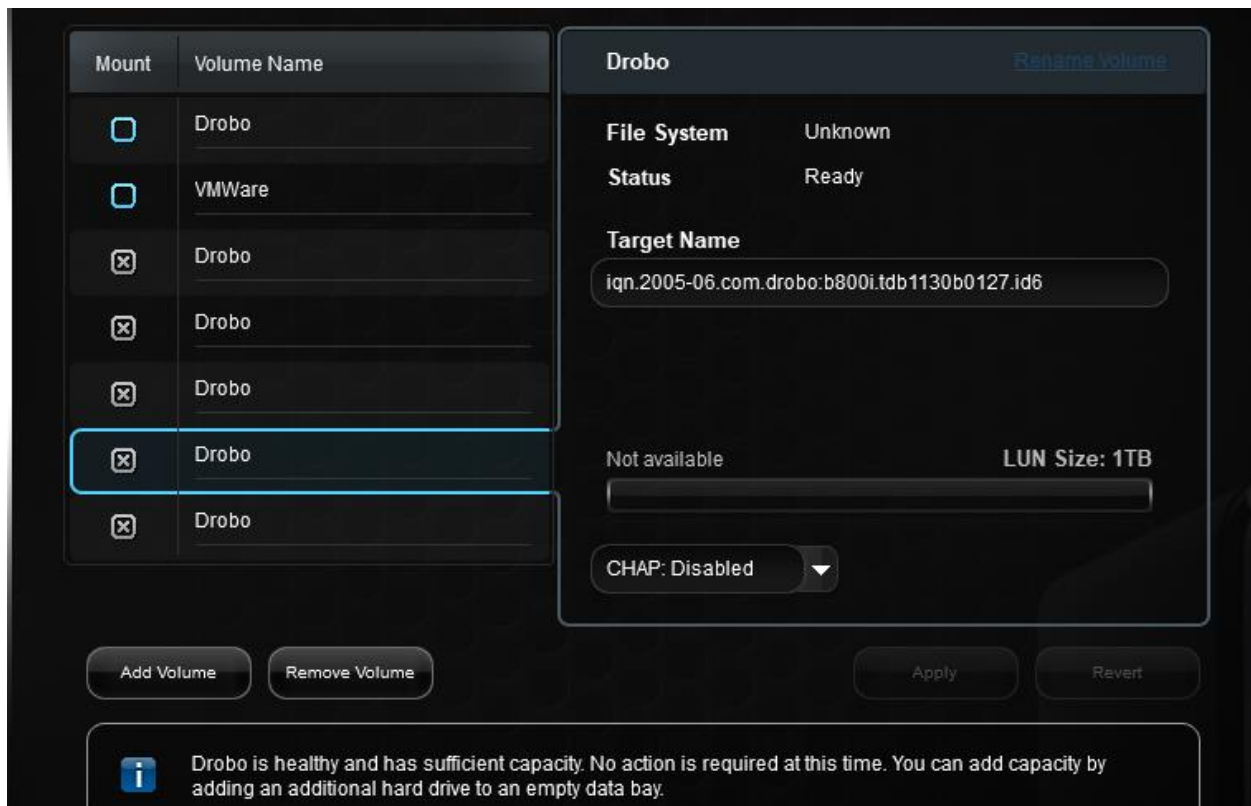
Category: Infrastructure & Security

Description: Provisioned a new CSI-Server on VMware ESXi and integrated Drobo NAS storage to support production workloads. Configured datastore creation, DNS mapping, and Apache2.

Process Documentation:

1. NAS Storage Setup

- Initialized new 1 TB disk in Drobo NAS.
- Created storage pool and volume, exported via iSCSI.
- Verified LUN mapping in Drobo Dashboard.



2. VMware ESXi Integration

- Connected ESXi host to NAS via iSCSI adapter.
- Created new datastore for CSI-Server VM.
- Allocated VM resources (CPU, RAM, storage).

Storage Adapters

[Refresh](#) [Rescan All](#)

Device	Type	WWN
iSCSI Software Adapter		
 vmhba37	iSCSI	iqn.2005-06.com.drobo:b800i.tdb1130b0127id2:D...
Patsburg 6 Port SATA AHCI Controller		
 vmhba0	Block SCSI	
 vmhba32	Block SCSI	
 vmhba33	Block SCSI	
 vmhba34	Block SCSI	

Details

vmhba37		Properties
Model:	iSCSI Software Adapter	
iSCSI Name:	iqn.2005-06.com.drobo:b800i.tdb1130b0127id2	
iSCSI Alias:	Drobo	
Connected Targets:	5	Devices: 5 Paths: 5

View: [Devices](#) [Paths](#)

Name	Identifier	Runtime Name
Drobo iSCSIDisk (naa.6001a620000642313133304230313237)	naa.6001a620000642313133304230313237	vmhba37:C0:T3:L0
Drobo iSCSIDisk (naa.6001a620000542313133304230313237)	naa.6001a620000542313133304230313237	vmhba37:C0:T2:L0
Drobo iSCSIDisk (naa.6001a620000242313133304230313237)	naa.6001a620000242313133304230313237	vmhba37:C0:T1:L0
Drobo iSCSIDisk (naa.6001a620000442313133304230313237)	naa.6001a620000442313133304230313237	vmhba37:C0:T0:L0
Drobo iSCSIDisk (naa.6001a620000742313133304230313237)	naa.6001a620000742313133304230313237	vmhba37:C0:T4:L0

CSI-Server

Getting StartedSummaryResource AllocationPerformanceEventsConsolePermissions

General

Guest OS:Ubuntu Linux (64-bit)

VM Version:7

CPU:4 vCPU

Memory:8192 MB

Memory Overhead:199.23 MB

VMware Tools:Unmanaged

IP Addresses:192.168.0.3

DNS Name:csi-server

State:Powered On

Host:lkphilsvm.lkphils.com

Active Tasks:

[View all](#)

Commands

 Shut Down Guest

 Suspend

 Restart Guest

 Edit Settings

 Open Console

Annotations

Notes:

 [Edit](#)

Resources

Consumed Host CPU:62 MHz

Consumed Host Memory:4148.00 MB

Active Guest Memory:81.00 MB

[Refresh Storage Usage](#)

Provisioned Storage:520.00 GB

Not-shared Storage:520.00 GB

Used Storage:520.00 GB

Datstore	Capacity	Free	Last Update
 csi-server	1,023.75 G	511.18 GB	2/6/2008 3
 datastore1	925.25 GB	385.11 GB	2/6/2008 3

Network

Type

 VM NetworkStandard switch network

View: [Datastores](#) [Devices](#)

Datastores

[Refresh](#) [Delete](#) [Add Storage...](#) [Rescan All](#)

Identification	Device	Capacity	Free	Type	Last Update	Hardware Acceleration
 csi-server	Drobo iSCSIDisk ...	1,023.75 G	511.18 GB	vmfs3	2/6/2008 3:00:10 PM	Unknown
 datastore1	Local VM Disk (v...	925.25 GB	385.11 GB	vmfs3	2/6/2008 3:00:10 PM	Unknown

3. Server Configuration

- Installed Ubuntu LTS on VM.

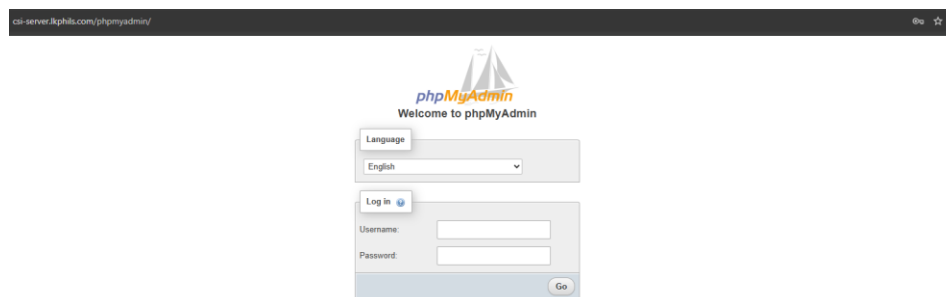
```
Resource Allocation Performance Events Console Permissions

mis@csi-server:~$ ls
mis@csi-server:~$ pwd
/home/mis
mis@csi-server:~$ cat /etc/os-release
PRETTY_NAME="Ubuntu 22.04.5 LTS"
NAME="Ubuntu"
VERSION_ID="22.04"
VERSION="22.04.5 LTS (Jammy Jellyfish)"
VERSION_CODENAME=jammy
ID=ubuntu
ID_LIKE=debian
HOME_URL="https://www.ubuntu.com/"
SUPPORT_URL="https://help.ubuntu.com/"
BUG_REPORT_URL="https://bugs.launchpad.net/ubuntu/"
PRIVACY_POLICY_URL="https://www.ubuntu.com/legal/terms-and-policies/privacy-policy"
UBUNTU_CODENAME=jammy
mis@csi-server:~$
```

- Configured Apache2, MySQL, phpMyAdmin.
- Created DNS entry `csi-server.lkphils.com`.

4. Outcome

- Delivered secure, production-ready CSI-Server.
- Enabled scalable storage via NAS integration.



Highlights:

- Configured Drobo NAS storage and integrated with VMware ESXi.
- Provisioned CSI-Server VM
- Managed DNS via firewall, Apache2 configs.
- Delivered production-ready environment for internal IT systems.